



University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

Project 3: University Examination Management and Grading System

Project Summary:

Build a complete **University Examination Management and Grading System** using C++ that simulates how a university department conducts exams, records marks, calculates grades, manages student records, and generates final result summaries. This program will make use of **functions** to structure the workflow, **file handling** for persistent record storage, **pointers** for efficient memory access to dynamic student data, and **2D arrays** to manage marks for multiple students and subjects.

Student & Subject Registration:

The administrator should first register all students who are appearing in the exam. For each student, store **Student Roll Number, Full Name, Program (e.g., BSCS, BSSE), Semester, and Section**. Use a **pointer array** to store student names dynamically, and make sure that each roll number is unique. Then, enter the list of subjects (e.g., Programming Fundamentals, Calculus, English) and store them as a 1D array of strings.

Exam Marks Entry Using 2D Arrays:

Allow the user to enter marks for each student in each subject. Use a **2D array** where each row represents a student and each column represents a subject. Marks must be validated between 0 and 100. The system should support up to **30 students and 6 subjects**. Input should be user-friendly and allow both batch entry and editing of specific student marks.

Grade Calculation and Result Sheet Generation:

Each student's percentage should be calculated based on the total marks obtained across all subjects. Then assign a grade using the following rules:

90–100% → A+

80–89% → A

70–79% → B

60–69% → C

50–59% → D

<50% → F

Grades should be stored in a separate 1D array. The system must also provide a **function to generate result cards** for each student showing detailed subject-wise marks, percentage, and grade.

File Saving and Searching:

Student marks, subject names, and final results should be saved in structured text files like marks.txt, students.txt, and results.txt. Use **file handling functions** to load existing records, update data, and append new entries. Also implement search functions to find a student by roll number and show their result card. Support should be included for reloading saved files when the program starts.

Required Functionalities Using Functions:

Use the following modular functions in your program:

addStudentDetails()

addSubjects()

enterMarks()

calculatePercentageAndGrade()

generateResultCard()



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saveToFile() and loadFromFile()
searchStudentResult()

Make use of **pass-by-reference** and **pointers** to process student records and dynamically access mark arrays.

Program Flow and Validations:

- Allow maximum 30 students and 6 subjects
- Roll numbers must be unique
- Marks must be numeric and in valid range
- Provide option to re-enter or correct a student's marks
- Show overall class statistics like highest and average marks

Use **nested loops** for subject-wise mark entry, and **if-else-if** for grade classification. Make your interface menu-driven.

File Structure Suggestions:

- students.txt → Roll No, Name, Program, Semester
- marks.txt → Student-wise marks
- results.txt → Roll No, Percentage, Grade

These should be saved in plain text format and easily readable.

Sample Input:

Register Student:

Roll No: 23F-BSCS-101

Name: Areeba Khan

Program: BSCS

Semester: 1

Section: B

Enter Marks:

Subjects: Programming, English, Calculus

Marks: 90, 85, 75

Search Student Result:

Roll No: 23F-BSCS-101

Sample Output:

Student added successfully.

Marks entered and stored.

Result for Areeba Khan (23F-BSCS-101):

Programming: 90

English: 85

Calculus: 75

Total: 250/300

Percentage: 83.33%

Grade: A